

COSC 290 Discrete Structures | Fall 2026 | Lab 3

Welcome to our third lab!

If you are not sitting next to someone, please move so that you are. After you and the person(s) who you are working with have checked your answer, you are welcome to continue working together if you'd like.

Collaborative question. Let's prove that:

Claim. Let $n \geq 1$. Then the integer $n^3 - n$ is always a multiple of 6.

- (a) Check the claim for the cases $n = 1, 2, 3$. One or more of these will be our base case.

- (b) The proposition $P(n)$ can be the statement in the claim. What is the induction assumption? It would start: We assume that $P(k)$ is true for.. (you finish the statement).

- (c) Now we will prove that $P(n + 1)$ is true. What, in words, is $P(n + 1)$?

- (d) We will need to do some algebra in order to convert the math in $P(n + 1)$ to expose the math in $P(n)$. Expand the binomial, collect like terms, and get the math in $P(n)$ isolated from the other terms.

- (e) If you have not already factored the terms that are not part of the induction assumption, then do so to get $3n(n + 1)$. Can you figure out why such an integer is always a multiple of 6?



← That's all we need! (Can you imagine writing it formally? If not, ask us.)

Feel free to continue working together!

1. I just want to point out that it is kind of cool that, in the collaborative question, one part of the algebra you encountered obeyed the induction assumption, while the other part required you to think about how integers work, and had nothing to do with induction.

That second part might inspire you to try sidestepping induction entirely. Try simply factoring $n^3 - n$ as far as possible, and then see if what you get inspires you in a way similar to how you had to think about the non-induction part of the collaborative question.



← Do you see how this approach avoids induction altogether?

2. Here is a curious solitaire game: Make a stack of n checkers. Split the stack into two in any way that you like, and earn points equal to the product of the sizes of those stacks. For example, if you start with $n = 8$ checkers, and split them into stacks of size 3 and 5, then you score 15. Keep doing this until all of the stacks are size 1. What is the best score you can achieve?

(a) Try to come up with a good strategy for maximizing your score.



← Tell us what you discover.

- (b) In this checkers game, the goal manifests itself directly: our first move is to split the stack of n checkers in two. Each stack will necessarily be smaller, and each itself represents a new checkers game. And in an induction proof, our goal is to reduce a problem to smaller but identical problems. So let's prove your discovery using induction!

First, what is your proposition $P(n)$?



← You should check in with us at each stage of this process.

(c) What is the base case?

(d) Suppose that we split the original stack of n checkers into stacks of sizes j and $n - j$. What is our score, and what will the scores of each of the two resulting games be?

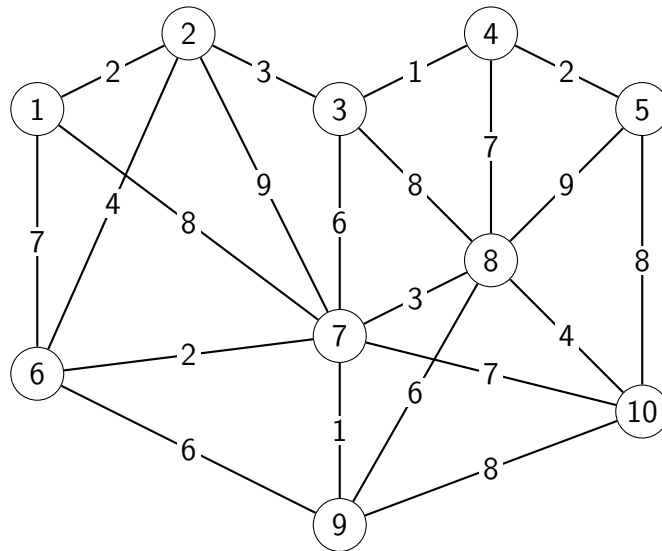
(e) Let's assume that $P(k)$ is true for $2 \leq k \leq n$, and prove that $P(n + 1)$ is true. What equation are we therefore trying to prove?

(f) Prove it! (Remember to pick one side and show, step by step, that it equals the other side.)

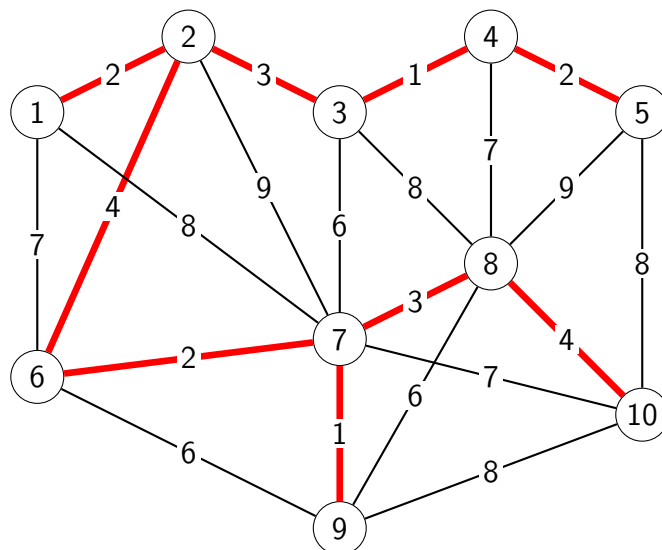


← Now we turn to an important proof in CS..

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3. Many algorithms iterate through a sequence of steps, and therefore we often prove the correctness of the algorithm using induction. Prim's algorithm, for example, finds a *minimum spanning tree* in an weighted undirected graph such as this:



A *cycle* in a graph is a path that starts and ends at the same node without revisiting any other nodes; for example, $7 \rightarrow 8 \rightarrow 10 \rightarrow 9 \rightarrow 7$ is a cycle. A *tree* is a subset of the edges that contain no cycles. A *spanning tree* reaches all nodes. A *minimum spanning tree* (abbreviated MST) is one with minimal total weight. Here is an MST for the graph above; the edges are red in the digital version on Discord, if that helps:



You may have studied Prim's algorithm in a different 200-level course. In case you have not, here is how the algorithm builds an MST:

Prim's algorithm (finds a minimum spanning tree T)

1. Choose any node to include in T .
2. Find an edge of minimum weight that connects a node in T to a node not in T .
3. Add that edge (and the node not yet in T) to T .
4. If all nodes are in T , stop; otherwise, loop to step 2.

(a) Try the algorithm on the graph above. Do you get the same MST that I did?

- (b) Prim's algorithm is straightforward, and clearly is aimed at finding an MST, but we need more than common sense in order to guarantee that an algorithm is correct. That's where a proof by induction comes in!

We cannot let our proposition $P(n)$ be "Prim's algorithm correctly finds an MST in a graph that has n nodes." Can you see why?

- (c) Instead, we let n be the number of steps that Prim's algorithm has taken thus far, as if we freeze time; in other words, we let n count how many nodes are in T . With this in mind, what is the base case?

- (d) We need to be a bit careful with how we phrase the proposition $P(n)$. If we say, "Let $P(n)$ be the proposition that after n steps, Prim's algorithm has created a minimum spanning tree," that would be incorrect. Why?

- (e) So instead, we let $P(n)$ say that after n nodes have been added by the algorithm, the algorithm is *on its way* to finding an MST. Here's how we put it:

Let $P(n)$ be the proposition that, after n nodes have been added by Prim's algorithm, the tree T is a subgraph of a minimum spanning tree.

Does your base case satisfy this proposition?

- (f) What would $P(n + 1)$ be?



← If you aren't checking with us at each part, check what you have so far.

- (g) Our setup is complete. Now, the interesting part! When you aren't quite sure how the logic part of an induction proof should go, you could try building up from the base case, so that you are considering something specific. Here, the base case is a single node u , so Prim's algorithm connects u to its neighbor v such that the weight of edge $e = (u, v)$ is minimum.

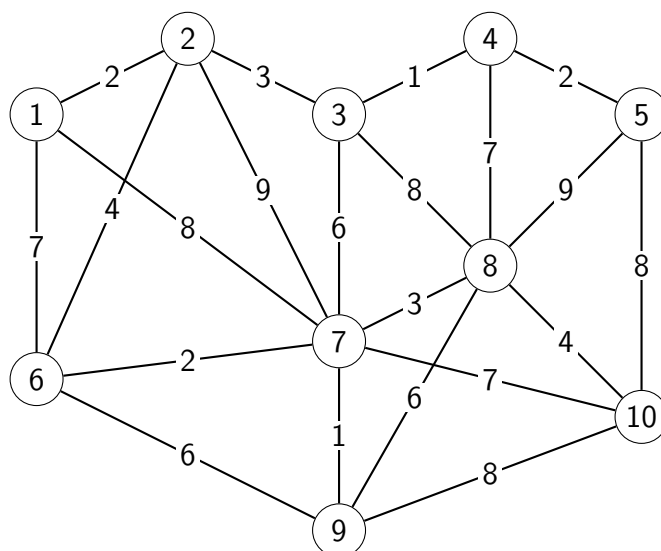
What question should we be asking ourselves about e ?

- (h) Well, if it is, then we are done, so we had better consider the case where it isn't, and prove that this case is impossible. Does that move make sense to you? It's like proving that it is raining by proving that it isn't not raining. Such a proof is called a proof by contradiction, and this one is sitting inside a larger induction proof.

So suppose that e is not an edge in an MST. Let M be an MST. If we were to add e (just for fun) to M , then M would no longer be a tree. Do you see why? If you aren't immediately sure, then you could look up at the specific MST and add any edge to it.

- (i) That cycle must contain two edges connected to u ; one is $e = (u, v)$, and let's call the other edge $e' = (u, w)$. Prim's algorithm always selects the edge of minimum weight when it adds a node, so it is not possible for the weight of e' to be greater than the weight of e . We are assuming that e is not an edge in an MST, but can we just remove e' from its MST and add e in its place, and if so, then is that a contradiction?
- (j) So we now have one edge that belongs to an MST. Do you think that the same argument that got us this edge can be applied again to get us a second edge?

- (k) If so, then all that remains is to say that if we ever have n edges in our tree that belongs to an MST, adding the next edge via Prim's algorithm will give us $n + 1$ edges that belong to an MST. The fundamental argument will not change. To solidify this point, here's that graph again:



Choose any node, and run Prim's algorithm for a few steps. Stop whenever you want (but before you reach every node). Now ponder the proof by contradiction above, and see if you can identify any logical problems that you aren't sure that you follow.



← That's a proof by induction that matters!

4. In class yesterday, I showed you what a *planar* graph is and how to count its *faces*.

- (a) Draw a couple of planar graphs (in their planar forms, so that no edges cross) and count their nodes, edges, and faces.
- (b) Let v be the number of nodes in a planar graph, and e be the number of edges, and f be the number of faces. These three numbers satisfy a certain equation. Rather than just tell you what it is, I would rather that you discover it. I think the best way to discover it is to imagine building a planar graph up from the base case. What would be v and e and f for a base case?

- (c) You have three variables to adjust as you build up from the base case. You can't add just a node, because then you would have two planar graphs, and our claim should be about just one planar graph. You can't add just an edge, because it wouldn't have a node at its far end. And you can't add just a face because a new face needs new edges.

So let's try adding just one edge and one node. This doesn't create a new face. This might make you think that the number of edges and the number of nodes might always increase together. But if you add another edge and node, it's possible to then add an edge without adding a new node. Are you following?

- (d) But that adds a face. So maybe adding an edge adds either a node *or* a face. Keep building onto your graph until you either agree or disagree with this. And do these explorations help you see how the three numbers always relate to each other?



← Look for these investigations to form the basis for Proof 2! See you tomorrow.